

Observable Patterns Are Not Explanations: A Causal-Geometric Analysis of Latent Reasoning Models

Darpan Aswal^{1,2} Thomas Palmeira Ferraz^{1,3} Yongxin Zhou¹ Maxime Peyrard¹

¹Université Grenoble Alpes, CNRS, Grenoble INP, LIG ²Université Paris-Saclay ³NAVER LABS Europe darpanaswal@gmail.com

EMNLP
2026
Main Conference



Code



Models

1 Motivation

Latent reasoning models (LRMs) replace explicit chain-of-thought with *continuous* latent thoughts, removing the visible trace used for monitoring.

Prior work reads observable latent-state patterns as evidence of reasoning: **BFS-like frontiers** in Coconut [1] and a decodable arithmetic **scratchpad** in CODI [2].

But a pattern can be **decodable without being causal**. We test *emergence* (matched controls), *generalizability* (two tasks, several LRMs) and *causality* (interventions).

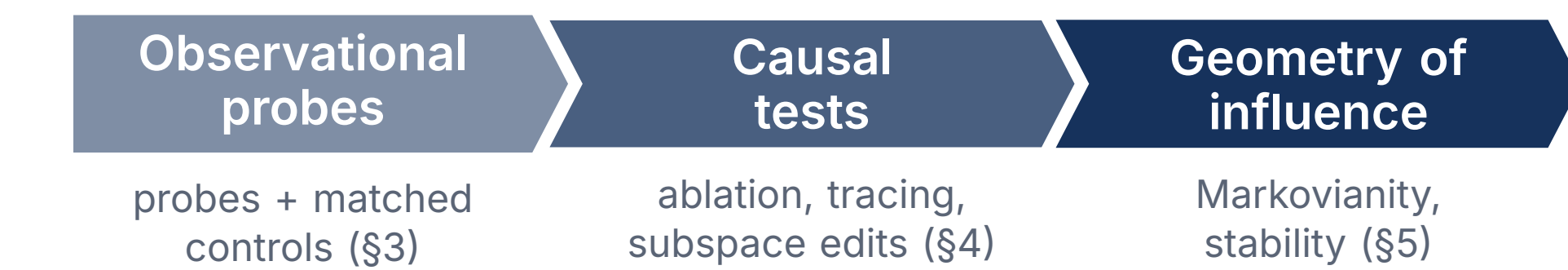
- RQ1** Do observable latent patterns uniquely emerge from latent-reasoning mechanisms and explain behavior? §3
- RQ2** When do latent thoughts influence predictions, and where is that influence concentrated? §4
- RQ3** Do behavior-influential thoughts evolve differently across steps from weakly influential ones? §5

2 Setup & Methods

Tasks graph-hopping (ProsQA), arithmetic reasoning (GSM8K).

Backbone GPT-2 small, trained per task (Llama-3.2-1B in the paper's appendix).

- C** LRM. Coconut [1]: latent recurrence, staged curriculum
- CODI** LRM. CODI [2]: latent recurrence, self-distillation
- PaT** **Recurrence control**. Pause-as-Thought: Coconut's curriculum, learned pause tokens, no recurrence [3]
- C_u** **Curriculum control**. Coconut that samples other curriculum stages with prob. $u=0.3$
- B** **Observational control**. Base GPT-2
- CoT** **Observational control**. GPT-2 fine-tuned on explicit CoT

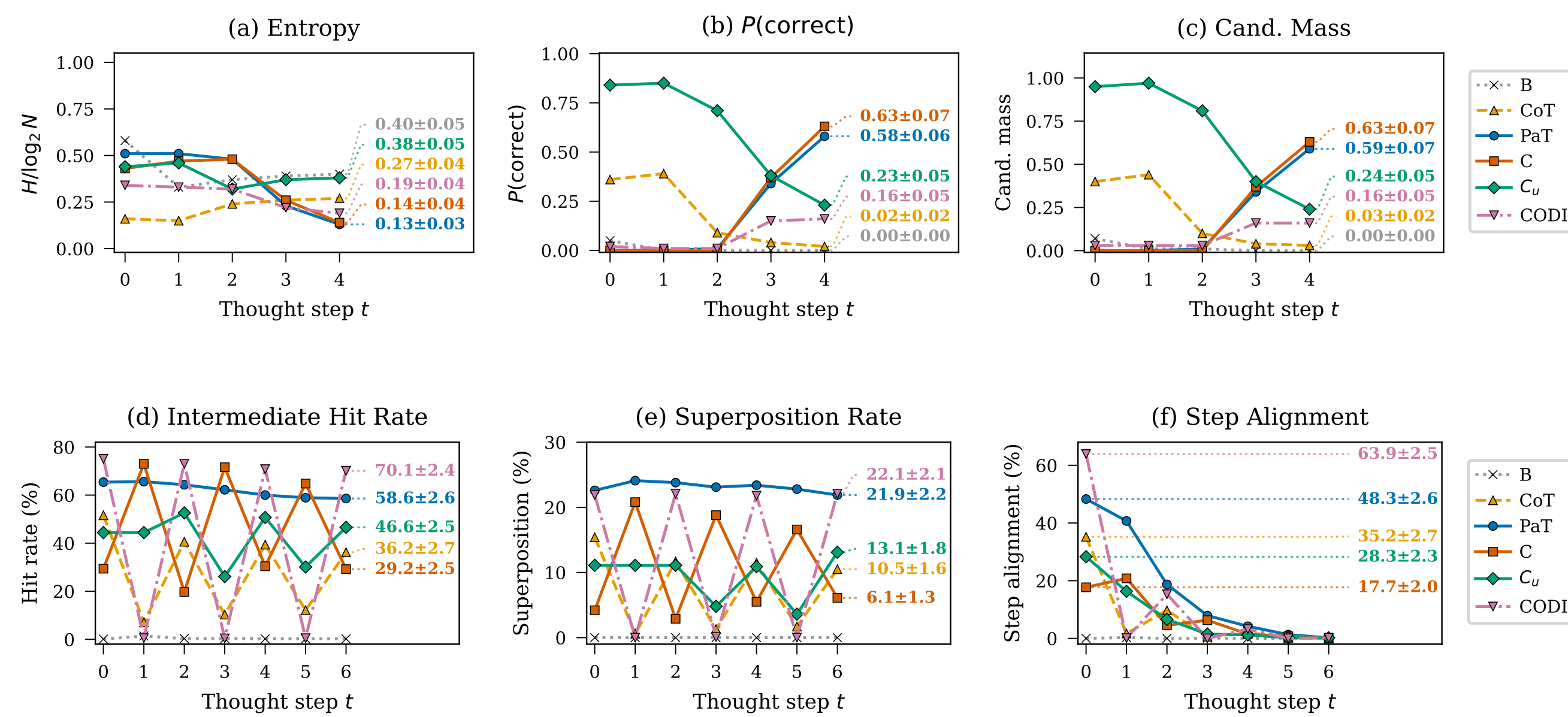


- Causal tracing [4]**. Patch a clean activation at site $s=(\ell, p)$ into a corrupted partner-prompt run: $\text{IE}(s) = 1 - \text{KL}_{\text{patch}}/\text{KL}_{\text{corr}}$ ($1 = \text{full recovery}$).
- Gradient subspace**. B_t = top right-singular vectors of $G_t = [\nabla_{h_{it}} \mathcal{L}_i]_i$ (95% energy). Edit $h_t \leftarrow h_t + (\alpha - 1) B_t B_t^\top h_t$; $\alpha=0$ ablates, $\alpha>1$ amplifies. Control: rank-matched random basis.
- Markovianity**. Predict h_t from $h_{t-1..t-n}$ with identity, mean, linear or MLP maps; test R^2 .
- Subspace stability**. $s_t = \text{mean } \sigma(B_t^\top B_{t+1})^2 \in [0, 1]$.

95% bootstrap CIs (1,000 resamples); McNemar tests for paired comparisons.

3 Observable Patterns Are Not Enough

We re-run the probes behind two published claims, adding controls that lack the proposed recurrence or curriculum.



Takeaway. Matched controls reproduce or invert the same patterns, so observable patterns are not sufficient for mechanistic attribution.

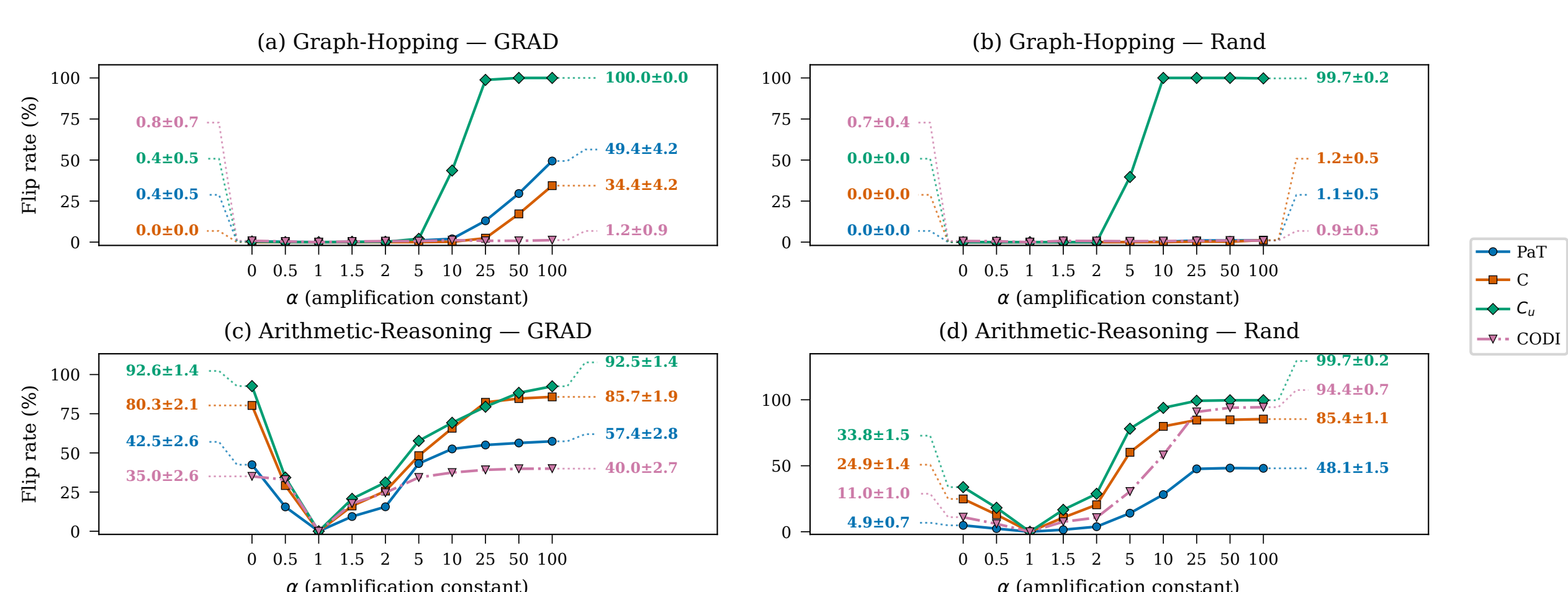
4 When and Where Do Latent Thoughts Matter?

Test-time thought ablation ($K \rightarrow 0$)

Model	Graph-hopping		Arithmetic	
	Full	Removed	Full	Removed
B	2.4 $\pm$ 1.3	—	1.4 $\pm$ 0.6	—
CoT	79.0 $\pm$ 3.6	—	41.9 $\pm$ 2.7	—
PaT	95.4 $\pm$ 1.8	95.6 $\pm$ 1.7	26.4 $\pm$ 2.4	21.4 $\pm$ 2.2
C	98.0 $\pm$ 1.3	97.8 $\pm$ 1.3	35.7 $\pm$ 2.5	7.7 $\pm$ 1.4
C _u	96.0 $\pm$ 1.7	91.8 $\pm$ 2.4	30.8 $\pm$ 2.5	39.1 $\pm$ 2.6
CODI	80.0 $\pm$ 3.5	80.0 $\pm$ 3.5	41.8 $\pm$ 2.7	25.6 $\pm$ 2.4

Table 1. Accuracy (%) with all latent thoughts vs. thoughts removed at test time.

Graph-hopping is barely affected. On arithmetic, **C** drops 35.7 $\rightarrow$ 7.7 and **CODI** 41.8 $\rightarrow$ 25.6, yet **C_u** improves: necessity is task-dependent, and **C_u** breaks the used-or-bypassed dichotomy.



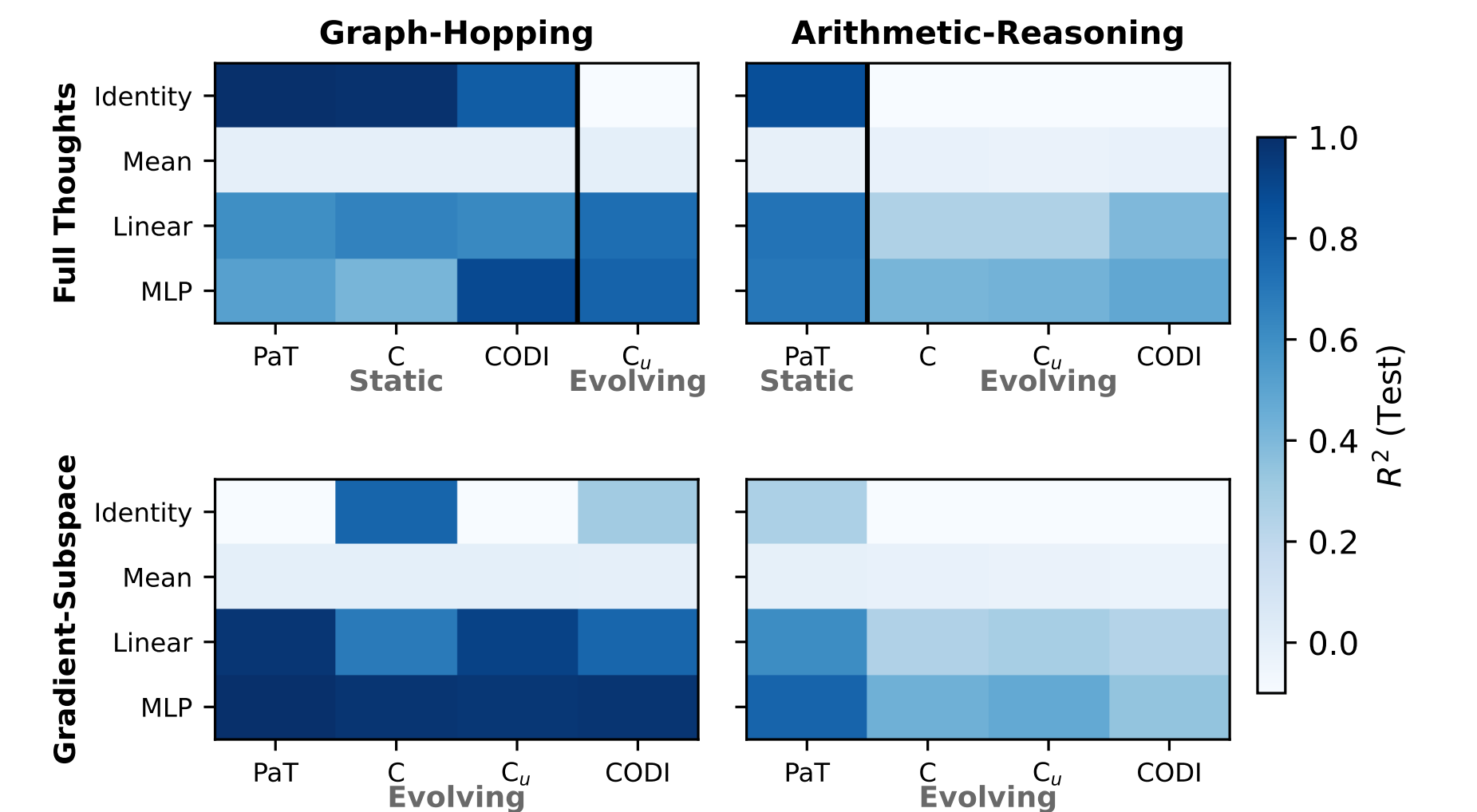
Takeaway. Latent-thought use is **graded**, not binary, and often concentrates in a low-dimensional, loss-sensitive subspace.

RQ1

5 Geometry of Influence

RQ3

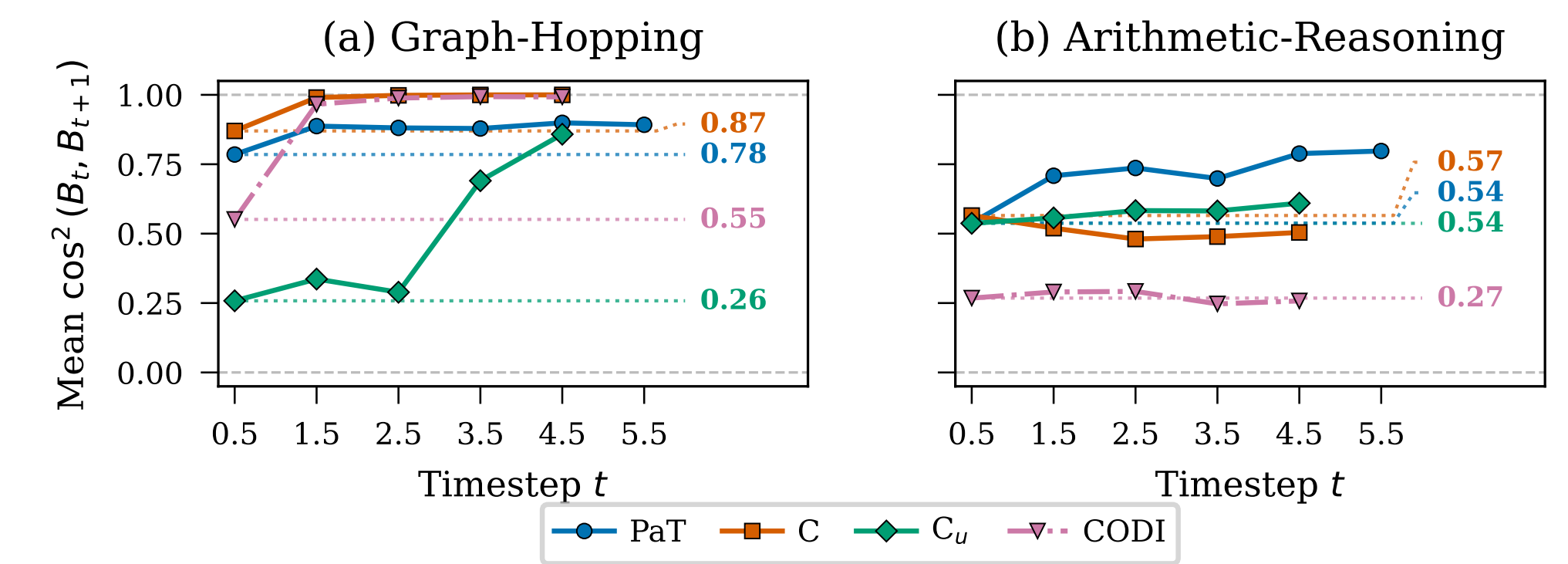
Markovianity of thought trajectories



Test R^2 of transition maps on full thoughts and on gradient-subspace projections $B_t B_t^\top h_t$.

- Full thoughts**. Graph-hopping: copying the previous thought matches any fitted map (static), except for **C_u**. Arithmetic: identity fails for **C**, **C_u**, **CODI** (evolving).
- Gradient subspace**. Every model evolves, even where full thoughts look static.

Stability of the gradient subspace



- Graph-hopping**: stable for **PaT**, **C**, **CODI**; **C_u** rotates early.
- Arithmetic**: lower stability; **CODI** rotates most.

Takeaway. Weakly influential thoughts are near-static; stronger influence comes with more structured subspace dynamics.

6 Takeaways

- Observable patterns are not explanations: they also appear in matched controls.
- Mechanism claims need matched controls *and* causal tests.
- Thought use is graded and task-dependent, concentrated in low-dimensional subspaces.
- Study the geometry of causally influential directions, not full latent states.

Latent thoughts are hidden computation, not hidden explanation.

[1] Hao et al., COLM 2025 (Coconut). [2] Shen et al., EMNLP 2025 (CODI). [3] Goyal et al., ICLR 2024 (pause tokens). [4] Meng et al., NeurIPS 2022 (causal tracing).